Accelerating Large-Scale Decision Tree Induction on Heterogeneous CPU-GPU Architectures

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Abstract. We investigate parallel implementations of two widely used decision tree induction algorithms, CART and C4.5, on multicore CPUs, CUDA-enabled GPUs, and heterogeneous CPU-GPU architectures.

Keywords: Decision trees · CART · C4.5 · Parallel computing

1 Introduction

2 Related Work

2.1 Decision Tree Induction: CART and C4.5

[Brief: CART [2] (Gini, binary splits, cost-complexity pruning), C4.5 [5] (gain ratio, MDL threshold cost, error-based pruning). Only what is needed to read Sect. 3.]

2.2 Scalable and Parallel Decision Tree Induction

2.2.1 Parallel Induction on Multicore CPUs

2.2.2 GPU-Accelerated Decision Tree Induction

3 Our Implementation and Optimizations

3.1 Serial Simple

3.2 Serial Optimized

3.3 Multicore CPU

3.4 GPU-Based

3.5 Hybrid

4 Experiment Design

4.1 Datasets

[One sentence per dataset (domain, how the test set is defined). All features numeric; ECBDL's nominal features as integer codes. The GPUs train ECBDL on its first 15M samples and SOREL on its first 4M (about 9.5G values each, 73 GiB of the 80 GB VRAM).]

Table 1. Datasets. Sizes are of the training set as float32. Every test set is the one defined with the data: the last 500,000 samples for SUSY and HIGGS (as the UCI page prescribes), the official test sets of ECBDL'14 and SOREL-20M, and for MNIST8M, which is generated from the MNIST training set, the 10,000-image MNIST test set. GPU subsamples are the first samples of the training set and use the full test set.

Dataset	Features	Classes	Training samples	Test samples	Size
SUSY [1]	18	2	4,500,000	500,000	0.3 GB
HIGGS [1]	28	2	10,500,000	500,000	1.2 GB
ECBDL'14 [6]	631	2	31,992,921	2,897,917	80.8 GB
GPU subsample	631	2	15,000,000	2,897,917	37.9 GB
MNIST8M [4]	784	10	8,100,000	10,000	25.4 GB
SOREL-20M [3]	2381	2	12,699,013	4,195,042	120.9 GB
GPU subsample	2381	2	4,000,000	4,195,042	38.1 GB

4.2 Hardware

[Four nodes of the mordor3 cluster (Table 2), each used whole (exclusive jobs). GPU nodes run the CUDA backend only, CPU nodes the CPU backends and the other CPU implementations. GPU runs are bound to the GPU's NUMA node (16 CPUs on hopper, 24 on ampere). Add measured memory bandwidth (STREAM) per node. Software: Ubuntu 24.04, Linux 7.0, g++ 13.3 (`-O3`), CUDA 12.9, NVIDIA driver 615.71.]

Table 2. Hardware. Every node has two sockets with SMT disabled; CPU, cores and L3 are per socket. Clock is base/max turbo in GHz. One GPU (PCIe) is used per run.

Node	CPU (per socket)	Cores	Clock	L3	RAM	GPU
hopper	Xeon Gold 6444Y	16	3.6/4.0	45 MB	1 TB DDR5-4800	H100 80 GB
ampere	EPYC 7402	24	2.8/3.35	128 MB	1 TB DDR4-3200	A100 80 GB
epyc1	EPYC 9255	24	3.25/4.3	128 MB	512 GB DDR5-6400	–
dell1	Xeon Gold 6542Y	24	2.9/4.1	60 MB	1 TB DDR5-5200	–

4.3 Compared Implementations

[./tree backends (`-simple`, `-serial`, `-parallel`, `-cuda`, `-gpu-only`); scikit-learn and Weka J48 on the CPU; cuML and XGBoost on the GPU, with their settings. Why each is the closest equivalent (no public GPU C4.5; J48 depth limit adapter; XGBoost squared error = Gini on bins). rpart and YaDT: add or say why not.]

4.4 Protocol and Measurement

[One tree, depth ≤ 10, no pruning, all samples and features; what is timed (training only, from data in memory; GPU setup and upload included); peak RAM and VRAM; runs per case, means and spread; exclusive nodes; fairness (CPU threads of the hybrid, warm library processes, input layout). Source: docs/GPU-LIBRARIES.md, bench/README.md.]

5 Results

5.1 Training Time

[Overview of every implementation on every dataset (Tables 3 and 4): speed-ups of Serial-Opt over Serial-Simple, Multicore over Serial-Opt, GPU over Multicore, and of this work over the existing implementations. Variance: report mean with min–max, or mean $\pm$ 95 % confidence interval, once every cell has 10 runs.]

5.2 Comparison with Other CPU Implementations

[./tree's backends against scikit-learn (CART) and Weka J48 (C4.5): time, memory, test accuracy, same trees.]

5.3 Comparison with GPU Libraries

[Against cuML and XGBoost: one tree, and per tree of a 10-tree forest; accuracy of exact versus binned trees; VRAM.]

5.4 Performance Across Hardware

[All ./tree backends on the four nodes, per dataset: serial, one and two sockets, H100 and A100 hybrid and GPU only. CPU against GPU; which hardware wins for which data shape (samples, features, classes).]

5.5 Scalability

[Thread scaling 1–48 on epyc1 and dell1 (memory bandwidth bound); growth with the number of samples (subsamples of one dataset).]

Table 3. Training time of one CART tree (depth ≤ 10, no pruning), in seconds: mean, and below it the fastest–slowest run. † marks a rough estimate (no run yet) with a plausible range below it; – not possible (data larger than the GPU's 80 GB, or no such algorithm). CPU runs on epyc1 (Multicore 48 threads, the rest one thread), GPU runs on the H100 (Hybrid with 16 CPU threads). Placeholder: every cell will have at least 10 runs.

This work					
Dataset	Serial-Simple	Serial-Opt	Multicore	GPU	Hybrid
SUSY	70.8 70.8–70.8	2.84 2.81–2.87	0.239 0.215–0.258	0.082 0.072–0.084	0.084 0.077–0.093
HIGGS	236 236–237	10.4 10.4–10.5	0.607 0.579–0.633	0.150 0.147–0.153	0.151 0.145–0.156
ECBDL'14	5 860	503 500–506	13.5 13.5–13.6	–	–
15M subsample	2 270	225 225–225	6.32 6.29–6.34	2.97 2.96–2.98	2.97 2.96–2.98
MNIST8M	1 880	234 233–235	5.92 5.90–5.94	3.57 3.54–3.57	3.56 3.54–3.57
SOREL-20M	12 500	878	21.8 21.2–22.2	–	–
4M subsample	2 920	234 234–234	6.43 6.40–6.48	2.87 2.85–2.91	2.87 2.82–3.02
Existing implementations					
Dataset	scikit-learn	rpart		cuML	XGBoost
SUSY	86.5 86.2–87.1	*140*† 90–250		0.133 0.126–0.147	1.43 1.42–1.45
HIGGS	215 214–215	*350*† 200–600		0.456 0.456–0.456	3.43 3.29–3.58
ECBDL'14	*4 500*† 3 000–9 000	*7 500*† 4 000–15 000		–	–
15M subsample	2 200	*3 600*† 2 000–6 000		9.95 9.90–10.0	42.2 42.1–42.4
MNIST8M	*2 500*† 1 500–5 000	*4 000*† 2 000–8 000		*7*† 4–14	*50*† 25–100
SOREL-20M	*13 000*† 7 000–26 000	*21 000*† 10 000–50 000		–	–
4M subsample	*3 500*† 2 000–7 000	*6 000*† 3 000–15 000		*10*† 5–20	*45*† 25–90

Table 4. Training time of one C4.5 tree (depth ≤ 10, no pruning), in seconds: mean, and below it the fastest–slowest run. † marks a rough estimate (no run yet) with a plausible range below it; – not possible (data larger than the GPU's 80 GB, or no such algorithm). CPU runs on epyc1 (Multicore 48 threads, the rest one thread), GPU runs on the H100 (Hybrid with 16 CPU threads). Placeholder: every cell will have at least 10 runs. cuML grows an entropy tree, its closest equivalent.

This work					
Dataset	Serial-Simple	Serial-Opt	Multicore	GPU	Hybrid
SUSY	77.2 77.1–77.2	3.10 3.06–3.12	0.249 0.221–0.308	0.109 0.094–0.114	0.112 0.108–0.116
HIGGS	257 257–257	11.9 11.9–12.0	0.590 0.568–0.628	0.164 0.159–0.171	0.170 0.167–0.172
ECBDL'14	6 890	623	17.6 17.5–17.7	–	–
15M subsample	2 590	277 277–277	7.81 7.76–7.86	2.99 2.99–3.00	2.99 2.99–3.00
MNIST8M	1 970	209 209–209	5.64 5.60–5.67	3.71 3.69–3.72	3.68 3.66–3.69
SOREL-20M	12 900	1 130	28.8 28.6–29.0	–	–
4M subsample	3 090	288 288–288	7.85 7.74–7.97	3.88 3.81–3.92	4.30 4.25–4.61

Existing implementations			
Dataset	Weka J48	YaDT	cuML
SUSY	230 229–230	*46*† 25–80	0.128 0.119–0.139
HIGGS	830	*170*† 90–300	0.433 0.414–0.457
ECBDL'14	*50 000*† 25 000–100 000	*10 000*† 5 000–20 000	–
15M subsample	24 200	*4 800*† 2 500–9 000	9.89 9.84–9.98
MNIST8M	*14 000*† 7 000–28 000	*2 700*† 1 200–6 500	*7*† 3–13
SOREL-20M	*60 000*† 30 000–120 000	*12 000*† 5 000–30 000	–
4M subsample	*18 000*† 9 000–40 000	*3 800*† 1 500–9 000	*10*† 5–20

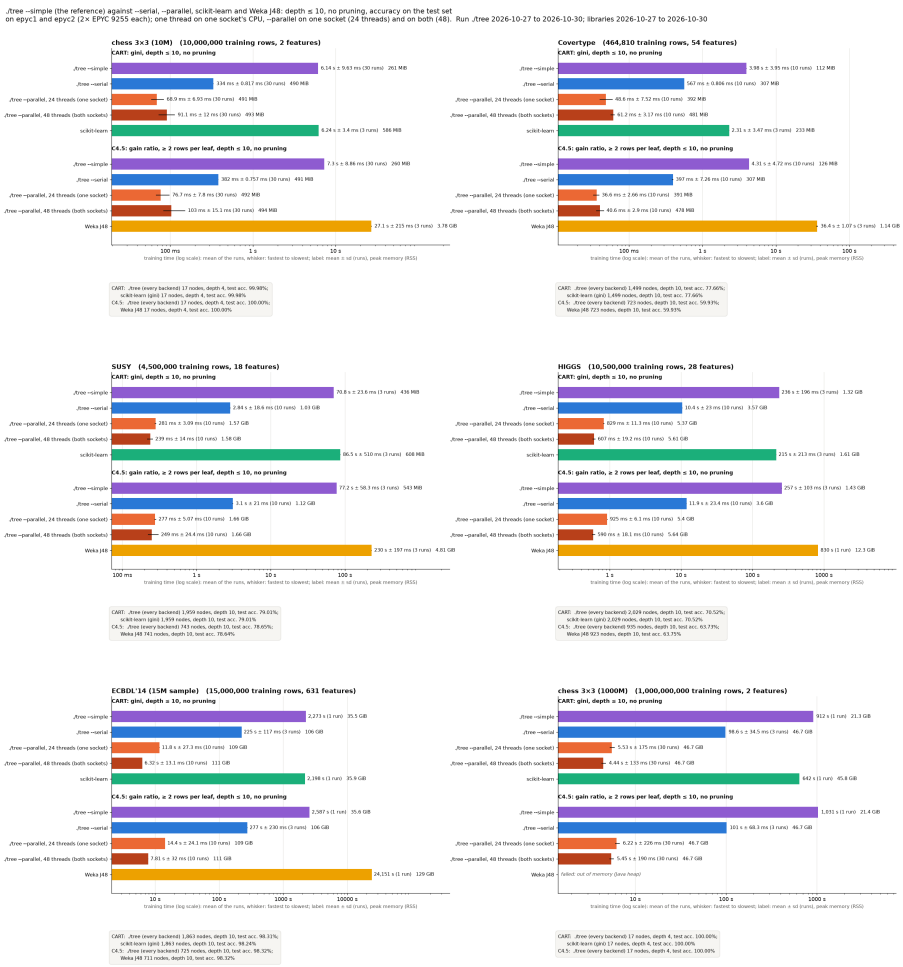


Fig. 1. Temporary: ./tree against scikit-learn and Weka J48 (cluster-cpu-vs-others).

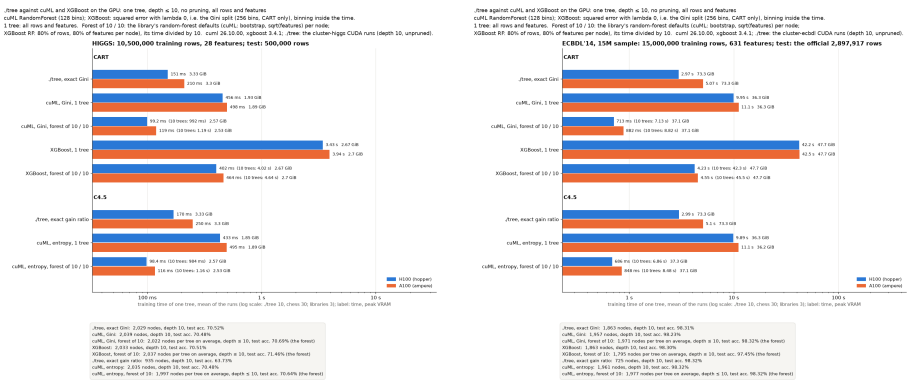


Fig. 2. Temporary: ./tree against cuML and XGBoost on HIGGS (left) and ECBDL'14 15M (right) (cluster-gpu-vs-others).

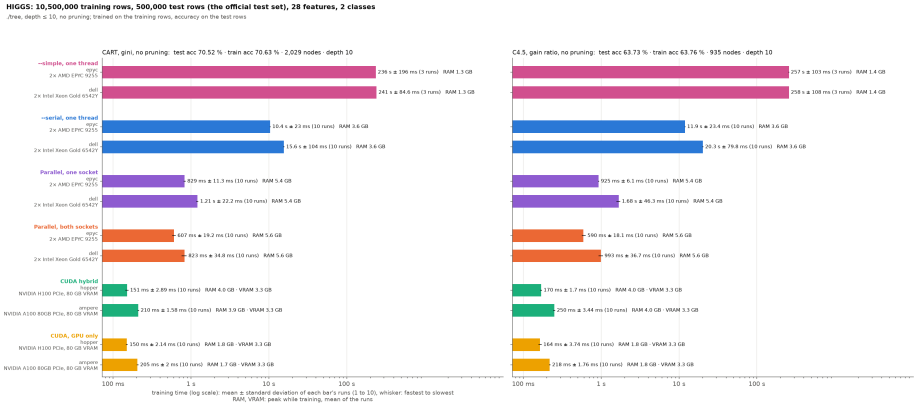


Fig. 3. Temporary: ./tree's backends on HIGGS on every node (cluster-higgs).

5.6 Profiling

[Where the time goes: presort, split search, partition, CPU–GPU transfer; GPU share of the hybrid; achieved bandwidth against peak.]

5.7 Energy Consumption

[Whole-server power while each implementation trains one tree on ECBDL'14 15M (the one dataset every implementation runs on, GPUs included). Method: power source and sampling rate (BMC/IPMI for the server, RAPL for the CPUs, `nvidia-smi` for the GPU), idle power measured before the runs and subtracted, energy = integral of power over the training time, 10 runs each. The CPU implementations run on a CPU node, the GPU ones on a GPU node, so the totals compare servers, not only processors. Discuss energy per tree: a fast run at high power can still use the least energy.]

Table 5. Energy of one tree on ECBDL'14 15M (depth $\leq$ 10, no pruning): mean training time t, mean power P of the whole server above idle, and energy $E = P \cdot t$. Placeholder: no measurements yet.

		CART			C4.5		
Implementation	Node	t (s)	P (W)	E (kJ)	t (s)	P (W)	E (kJ)
Serial-Simple	epyc1	?	?	?	?	?	?
Serial-Opt	epyc1	?	?	?	?	?	?
Multicore	epyc1	?	?	?	?	?	?
GPU	hopper	?	?	?	?	?	?
Hybrid	hopper	?	?	?	?	?	?

6 TEMPORARY: Chart Style Tests

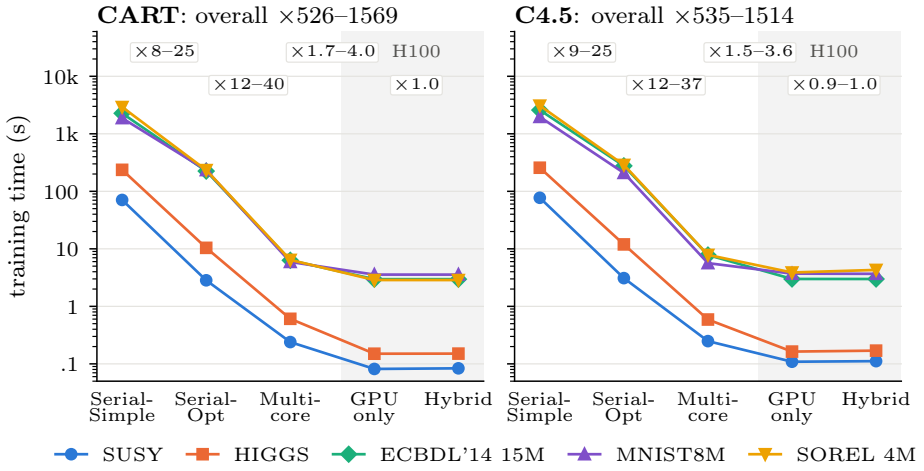


Fig. 4. Training time of one tree (depth $\leq$ 10, unpruned) through the steps of this work: Serial-Simple, Serial-Opt and Multicore on epyc1, then GPU and Hybrid on the H100 (grey). Boxes: range of the step's speed-up over the five datasets.

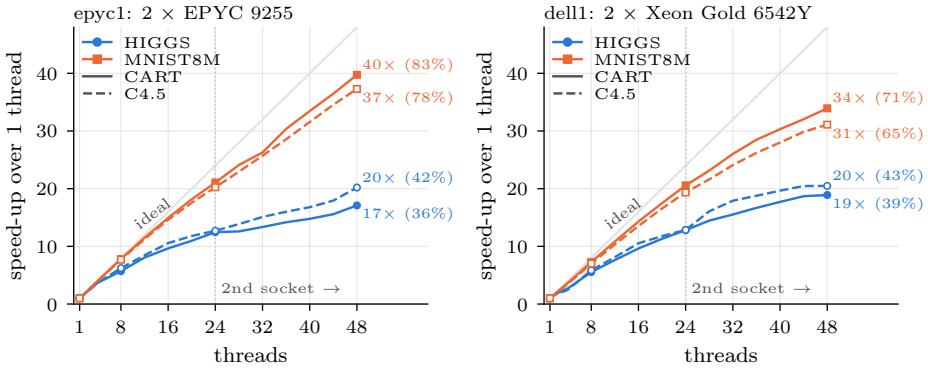


Fig. 5. Speed-up of Multicore over one thread on epyc1 (left) and dell1 (right), depth $\leq$ 10, unpruned. Labels: speed-up and parallel efficiency at 48 threads; dotted line: the second socket starts at 25 threads.

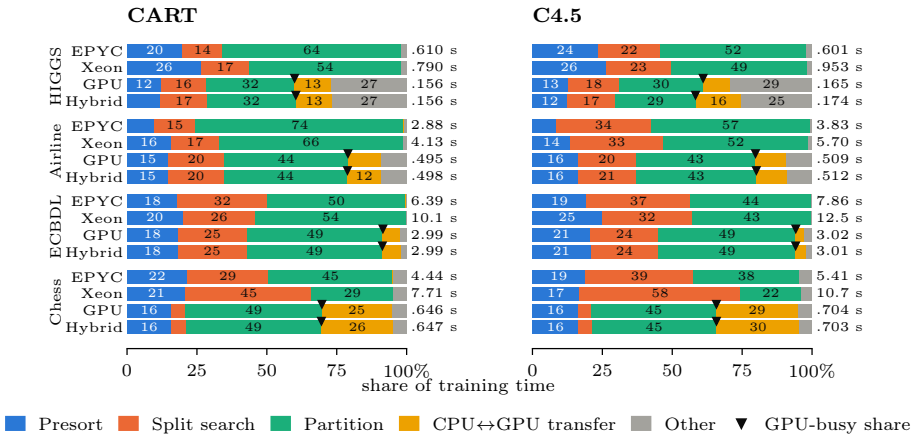


Fig. 6. Where the training time goes (percent of the training time; mean of the profiled runs, depth ≤ 10, unpruned). EPYC, Xeon: Multicore with 48 threads on epyc1 and dell1; GPU and Hybrid on the H100. Triangle: share of the time the GPU is busy. Chess: 3×3 board, 10^9 samples.

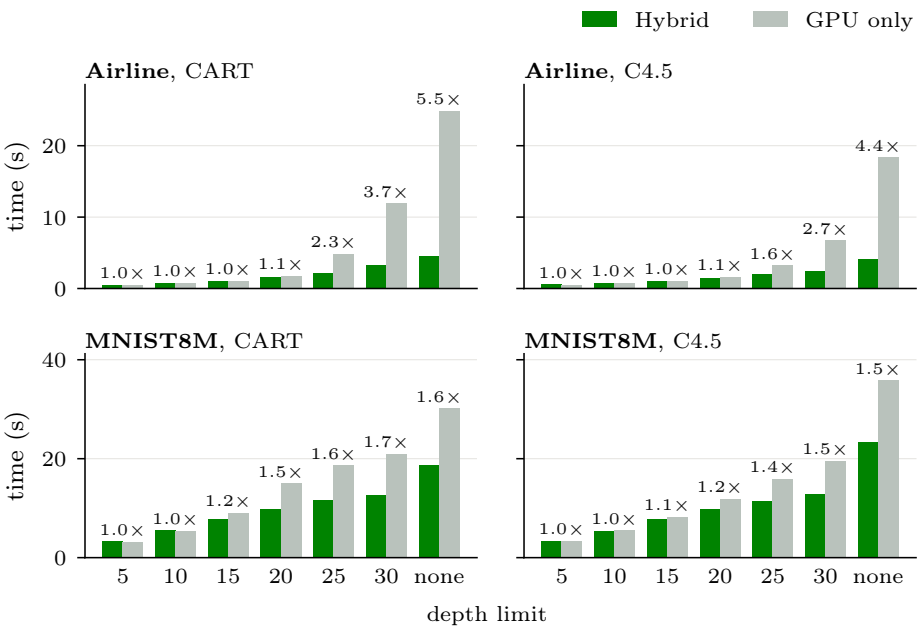


Fig. 7. Hybrid against GPU by depth limit on the A100 (unpruned, mean of 3 runs). Numbers above the bars: how many times slower GPU is. Both modes grow the same trees.

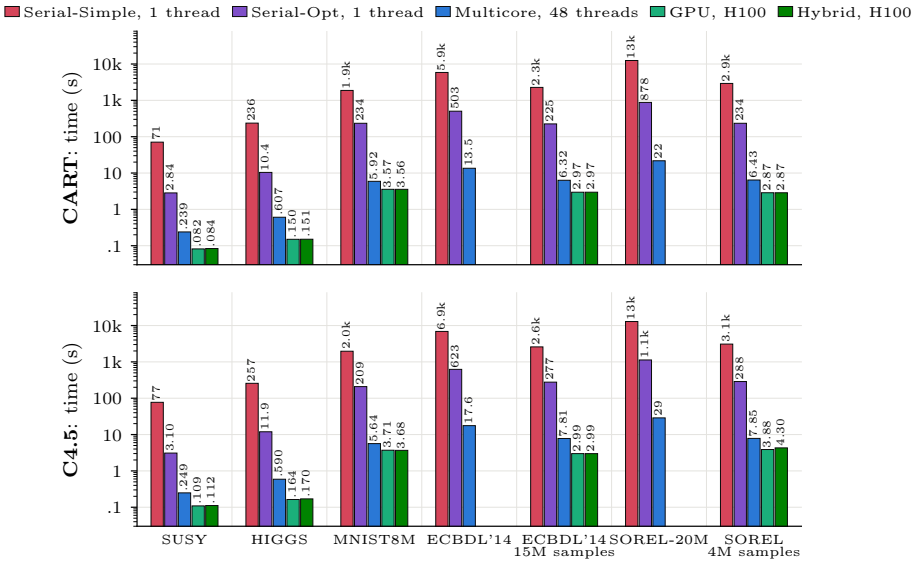


Fig. 8. Training time of one tree (depth $\leq$ 10, unpruned), log scale: mean over the runs. Serial-Simple, Serial-Opt and Multicore on epyc1, GPU and Hybrid on the H100 (Hybrid with 16 CPU threads). No bar: no run (the full ECBDL'14 and SOREL-20M do not fit in the 80 GB of VRAM).

7 Discussion

8 Conclusion

References

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